

Project Trishul - Frontend Completion Walkthrough

All core frontend components and pages planned for Phase 1 are fully built, tested, and verified.

☀️ What Has Been Built & Verified

1. 🏠 Public Landing Page (/ - Home.jsx)

- **Live Emergency Marquee & Threat Ticker:** Top bar alert system for regional emergencies.
- **Hero & IoT Telemetry Matrix:** Live telemetry counters (342 active sensors, high risk zones, response units, AI Antecedent Rain Index accuracy).
- **Interactive 8-State North East Threat Matrix:** Filter by threat level (`CRITICAL` , `HIGH` , `MODERATE` , `LOW`) with detailed sector focus drawers.
- **Live Verified Field Incident Bulletins:** Real-time disaster bulletins verified by field officers.
- **Key SIH Architectural Pillars:** Explains AI rainfall thresholds, zero-connectivity offline PWA, and graph resilience.

2. 📱 Citizen & Field Officer Incident Reporting Portal (/report - ReportIncident.jsx)

- **Offline-First PWA Sync Engine:** Detects connectivity changes; allows queuing incident reports in local storage during network blackout and syncing when back online.
- **One-Touch GPS Geolocation Acquisition:** High-accuracy coordinate fetching.
- **Multi-Hazard Categorization:** Landslides, road blockages, slope tension cracks, flash floods, culvert/bridge damage.
- **Geo-Tagged Evidence Photo Attachment:** Camera snapshot/file upload with instant preview.
- **Emergency SOS Distress Mode** (/report?sos=true): High-priority emergency broadcast.
- **Live Ground Truth Feed:** Real-time list of reports with timestamps and location tags.

3. 🗺️ Mountain Road Passes & Safe Evacuation Navigator (/safe-routes - SafeRoutes.jsx)

- **AI Evacuation & Safe Bypass Route Finder:** Calculates stable bypasses (e.g., Siliguri to Gangtok via Melli/Namchi) avoiding active river mudslides.
- **Strategic Highway Corridors Status Grid:** Live tracking of NH-10, NH-29, NH-37, BCT Road, NH-06, and NH-54.
- **Detailed Hazard Bottlenecks & BRO Clearance ETAs.**
- **Isolated Mountain Village Vulnerability Tracker:** Evaluates isolated communities (Lachung, Dzongu, Mahur), access lifelines, and emergency ration buffers.

4. 🛡️ Authority Command Center Dashboard (/admin - AdminDashboard.jsx)

- **Interactive Tactical GIS Map** (`TacticalGisMap.jsx`):
 - Multi-layer toggles for **Incidents**, **IoT Sensors**, and **Response Teams**.
 - 100% Free tile layers: OpenStreetMap Standard, Topographic Relief, and Esri Satellite Imagery.
 - Incident risk perimeter circles and interactive popup telemetry cards.
 - **AI Antecedent Rain Index vs. Soil Saturation Tipping Curve:** Interactive Recharts graph predicting slope collapse probability.
 - **Incident Triage & Verification Workflow:** Full lifecycle state transitions (`PENDING_REVIEW` → `VERIFIED` → `RESOLVED`).
 - **Geo-Fenced Emergency Broadcast Studio:** Multilingual emergency siren dispatcher (SMS / WebPush / Cell Broadcast).
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Navigation Summary

Route	Page	Purpose
/	Home.jsx	Public Threat Matrix & Regional Overview
/report	ReportIncident.jsx	Citizen Geotagged Incident Reporting & Offline Sync
/safe-routes	SafeRoutes.jsx	Highway Corridor Clearance & Safe Bypass Route Finder
/admin	AdminDashboard.jsx	Authority Command Center, GIS Map & Emergency Siren



Next Steps (When You Return from Your Break)

- **Phase 2 - Backend & Database:**
 - Initialize Node.js + Express backend inside `server/` .
 - Configure MongoDB Atlas with `2dsphere` geospatial indexing for live radius queries.
 - Set up REST API endpoints (`/api/incidents` , `/api/corridors` , `/api/weather`) and Socket.io for live updates.